MASTER CLINICAL SUPPLY QUALITY AGREEMENT (Q-AGR)

Document Ref: Moderna Deep-Frozen Distribution Standard (Q-AGR-MRNA-09) | **Standard:** FDA 21 CFR Part 11 & EU GMP Annex 15

1. Commercial Parties & Asset Specifications

Commercial Sponsor	Moderna Oncology Therapeutics
Sponsor Identifier	CUST-MRNA-03
Therapeutic Asset	**Custom Personalized mRNA Neoantigen Vaccine Lot #ONC-77**
Active Batch / Lot #	LOT-MRNA-NEO-077
Commercial Valuation	**$390,000 USD**
Consignee Medical Facility	MD Anderson Cancer Center Cellular Suite

2. Storage Specifications & Excursion Limits

The carrier is legally and operationally bound to preserve custody within qualified physical boundaries under Tier **Deep-Frozen mRNA Stability SLA (-20°C ± 5°C) (SLA-FROZEN-MKT)**. Continuous edge telemetry samples temperature at ≤ 1.0 second frequency with cryptographically verifiable records.

Qualified Temperature Band	**-25.0°C to -15.0°C**
Critical Regulatory Ceiling	-15.0°C
Critical Regulatory Floor	2.0°C
Allowable Cumulative Excursion	45 minutes maximum
Sampling & Seal Frequency	Real-Time Telematics (1.0 sample/s)

3. Financial Liability & Liquidated Damages Schedule

Failure to remediate an active physical breach beyond -15.0°C within 45 minutes constitutes complete batch degradation, triggering full commercial reimbursement of **$390,000 USD**. For delays exceeding contract cutoff without thermal breach, liquidated damages accrue at **$40,000 regulatory re-filing and cohort postponement penalty**.

4. Autonomous Triage & Emergency Diversion Protocols

Under FDA 21 CFR Part 11 electronic control provisions, the commercial sponsor confers autonomous executive authority to **CryoGuard AI™**. In the event of primary equipment failure, the system is authorized to:

• Perform multi-criteria facility scoring (distance, bay capacity, certified capabilities).
• Execute closed-loop highway diversion to the nearest compatible secondary biorepository.
• Automatically generate cryptographically signed CAPA non-conformance reports.

5. Regulatory Certification & Electronic Signatures

For Commercial Sponsor: Moderna Oncology Therapeutics *Digitally Signed: e-Sig SHA-256* Date: 2026-10-14	**For Logistics Carrier & AI Operator:** CryoGuard Autonomous Logistics Inc. *Automated Execution Authority Active* Date: 2026-10-14